# Governance-Aware Weak-Signal Architecture for Grid Anomaly Triage

A Bridge Note on Operator-Facing Early Warning Under Uncertainty

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## Abstract

Grid monitoring and energy-system anomaly handling often begin not with cleanly resolved events, but with weak, partial, and locally ambiguous signals: small voltage deviations, mild thermal irregularities, feeder imbalance, telemetry inconsistency, demand anomalies, or under-response from distributed energy resources. In such cases, the architectural problem is not only detection, but promotion control: how weak findings should influence attention before they are allowed to become stronger operational significance. This bridge note argues that operator-facing anomaly triage is a natural domain for a weak-signal architecture organized around four bounded stages: bounded findings, provisional candidate meanings, attentional escalation, and governed promotion. The note contributes a minimal contract surface for that transfer: what a bounded finding contains, what a candidate state contains, what kinds of escalation actions are permitted, and what conditions must be met before promotion occurs. It does not propose control algorithms, market mechanisms, dispatch logic, or certified operational deployment.

## 1. Problem Framing

Energy systems often produce early signals that are too structured to ignore but too weak to deserve immediate commitment. A transformer may show a mild thermal rise while nearby telemetry remains inconsistent. A feeder may display a local imbalance that differs meaningfully from recent conditions without yet supporting a strong equipment-failure claim. A demand anomaly during a stressed interval may reflect genuine localized strain, transient behavioral irregularity, or measurement fault. In each case, the system faces the same question: what should happen when the current pattern no longer fits the cheap local corridor, but stronger action still requires more support?

This is where a weak-signal architecture becomes useful. The relevant problem is not only whether an anomaly exists in some abstract sense. The problem is how early significance should be represented, escalated, inspected, and either promoted or allowed to decay. If the system ignores weak but informative departures because they do not yet cross a strong threshold, it loses useful lead time. If it treats provisional anomalies as settled truth too early, it creates brittle state, unnecessary operator burden, or misdirected intervention. The architecture therefore needs a disciplined middle state in which ambiguity can remain explicit while scope widens in bounded ways.

This paper is aimed at system architects, platform engineers, and operator-tool designers working on early-warning and triage layers rather than on final control. It is a bridge note, not a standards document and not a utility operations paper. Its purpose is to show that the weak-signal stack already used elsewhere transfers cleanly into grid anomaly triage when the transfer is stated in minimal operational terms.

## 2. Transfer Thesis

The transfer claim is narrow. Operator-facing grid anomaly triage is a natural domain for a weak-signal architecture in which bounded findings raise attention first, candidate meanings remain provisional, and promotion to stronger operational significance occurs only through declared persistence, corroboration, and governance predicates.

The architectural sequence is simple:

bounded finding → provisional candidate → attentional escalation → governed promotion

What transfers is not a claim about grid optimization or control. What transfers is a way of handling underdetermined significance. The same structural logic appears in other weak-signal settings: local departures matter before they settle meaning, repeated or convergent evidence widens scope, and stronger consequence requires more than short-horizon salience alone. In this energy setting, that means a weak pattern should first change what the system looks at next before it changes what the system is allowed to treat as stronger operational state.

## 3. Minimal Architecture Contract

This section is the technical core of the bridge note. The point is not to prescribe a full implementation. It is to define a minimal contract surface that makes the transfer concrete.

### 3.1 Bounded Finding

A bounded finding is a narrow, source-specific output produced by a local sensing or monitoring path. It is not a resolved anomaly claim and not a control instruction. At minimum, a bounded finding should contain: source or subsystem identity, timestamp, scoped location or asset reference, measured signal or metric, deviation relative to the relevant local baseline, confidence or reliability score, and provenance sufficient for replay or inspection. In energy terms, a bounded finding may be a small voltage sag relative to recent feeder behavior, a mild thermal rise on a transformer, a frequency irregularity, a local demand deviation, a telemetry inconsistency, or weak DER under-response relative to setpoint. The architectural rule is that bounded findings preserve local structure without claiming global meaning.

A minimal bounded finding could take a compact form such as: source_id = TR-42; timestamp = 2026-04-13T13:02:00; scope = Feeder-7 / Segment-A; metric = transformer_temp_C; observed_value = 88.3; delta_from_local_baseline = +6.2; confidence = 0.72; provenance =

telemetry_stream_v2 / msg12345. The point of such a record is not to resolve meaning, but to preserve a narrow, replayable local deviation for later interpretation.

### 3.2 Provisional Candidate State

A provisional candidate state is an interpretation object formed from one or more bounded findings. It represents admissible local meanings without pretending uncertainty has vanished. A candidate state should minimally include: hypothesis label, supporting findings, confidence or score, scope, and a short-lived decay condition or review horizon. Example candidate states in this domain include possible local congestion, possible equipment stress, possible unstable load pocket, possible telemetry fault, possible DER coordination issue, or unresolved anomaly requiring more evidence. Competing candidates may coexist; the architecture does not become stronger by hiding that competition.

A minimal provisional candidate could take a form such as: candidate_id = cand_017; hypothesis = possible_equipment_stress; supporting_findings = [TR-42_temp_rise, feeder_load_shape_shift]; scope = Feeder-7 / Segment-A; score = 0.68; review_horizon = 15 minutes; decay_rule = expire_if_no_new_support. This kind of record keeps the interpretation explicit, bounded, and revisable.

### 3.3 Attentional Escalation

Attentional escalation is the stage at which weak significance changes what the system is allowed to inspect or request next without yet becoming durable significance. Permitted escalation actions are bounded and reversible. In this domain they may include requesting additional telemetry, shortening the sampling horizon, increasing inspection priority, seeking corroboration across channels or assets, opening an operator-facing review band, or applying conservative temporary smoothing or watch-state logic at the triage layer. The important boundary is that escalation changes attention, evidence gathering, and local scrutiny; it does not itself authorize curtailment, dispatch, or irreversible operational action.

### 3.4 Governed Promotion

Governed promotion is the boundary at which a candidate may become a stronger anomaly state for operator-facing routing. Promotion requires more than mere salience. At minimum, promotion should be gated by declared predicates such as persistence across a bounded review window, corroboration from multiple findings or channels, meaningful scoped deviation relative to the relevant baseline, and a rationale record sufficient for audit.

Example promotion predicates might include: persistence across a declared review window; corroboration from at least two independent findings or channels within the same scope; or repeated reappearance of the same candidate state across bounded intervals with consistent supporting evidence. These are examples rather than fixed thresholds. The architectural

requirement is that promotion be declared, inspectable, and predicate-bound rather than silently triggered by salience alone.

A promotion record should preserve the candidate label, supporting evidence, relevant scopes, confidence score, and the reason the case crossed from provisional attention into stronger triage significance. Promotion may be performed by an automated triage gate, a human operator, or a hybrid review path, but in all cases the record should preserve who or what authorized the move, which predicates were satisfied, and what evidence supported the transition. If those predicates are not met, the candidate remains provisional, decays, or is quarantined for later inspection. That is what governance means in this note: explicit promotion control with inspectable rationale, not a vague appeal to caution.

## 4. Three Compact Decision Flows

The examples below use the same structure: inputs, candidate formation, escalation step, and governed outcome.

### Example 1: Transformer / Feeder Precursor

A mild transformer thermal rise appears together with an unusual local load shape and nearby telemetry inconsistency. The bounded findings are recorded separately with source, time, local delta, and confidence. The interpretation layer forms two candidate states: possible equipment stress and possible telemetry fault. Because the signal is meaningful but underdetermined, the system raises attention rather than promoting either candidate immediately. It requests additional telemetry, increases inspection priority, and places the asset in a bounded watch state. Promotion occurs only if the thermal rise persists across the declared review window or gains corroboration from additional channels; otherwise the case decays or remains tagged as inconclusive.

### Example 2: Demand-Response Ambiguity

During a stressed interval, a local load anomaly appears that could indicate genuine strain, transient demand irregularity, or measurement error. The system records the bounded finding relative to the scoped baseline for that period and forms candidate states for localized stress and measurement anomaly. Because neither candidate is yet strong enough, the escalation layer requests corroboration from adjacent telemetry and opens an operator-facing review band rather than treating the anomaly as settled. Promotion occurs only if persistence and cross-source support strengthen one candidate beyond the declared review conditions. If support does not build, the anomaly remains provisional and expires from the higher-priority queue.

### Example 3: DER / Microgrid Instability Precursor

A distributed resource cluster shows weak under-response together with intermittent local instability markers. The architecture records each as bounded findings with provenance and scope, then forms a provisional candidate for possible DER coordination issue and a competing

candidate for localized telemetry instability. The salience layer responds by tightening short-horizon monitoring and requesting corroboration from related nodes or devices. Governed promotion occurs only if the under-response persists, converges with additional instability evidence, or repeatedly reappears within the declared window. Until then, the case stays in triage rather than becoming a stronger operational claim.

## 5. Why This Architecture Matters

The value of this architecture is not that it eliminates uncertainty. The value is that it gives uncertainty a disciplined place to live before stronger commitment occurs. In grid anomaly triage, that middle state matters because weak early signals are often the only source of lead time, yet they rarely deserve immediate hardening into stronger operational significance.

This arrangement helps prevent two symmetric failures. The first is false indifference: weak but informative departures are ignored because they do not yet cross a strong threshold. The second is false certainty: provisional anomalies are promoted too early and begin to behave like settled truth in operator workflows. A stratified weak-signal architecture reduces both errors by separating local finding, provisional interpretation, attention management, and promotion control into distinct stages with distinct permissions.

Just as important, the architecture makes inspection easier. When a case is promoted, there is an explicit rationale trail showing which findings, scopes, and predicates supported the move. When a case is not promoted, the reason is also visible: lack of persistence, weak corroboration, or insufficient scoped deviation. That is the practical sense in which this bridge note is governance-aware.

## 6. Boundaries and Non-Goals

This note draws hard boundaries around what the proposed architecture is and is not.

This architecture does not perform automatic control actions. It may support operator-facing triage or recommendation pathways, but automatic actuation is outside the scope of this paper.

This note does not propose market design, settlement, bidding logic, or any mechanism that alters financial outcomes.

This architecture is not a certified control system and does not claim deployment readiness for automated intervention.

This note does not propose dispatch optimization, formal power-flow novelty, or full DER coordination logic. It is concerned only with early anomaly handling, salience management, and governed promotion of candidate anomaly states.

This paper assumes provenance-rich telemetry and an operator-facing workflow. Severely degraded telemetry environments would require additional instrumentation and are outside the present bridge note.

These limits matter because the transfer claim is narrower than the surrounding operational domain. The paper is not saying it solves grid operations. It is saying that operator-facing anomaly triage is another serious domain in which weak signals should matter before they become stronger state.

## 7. Conclusion

Grid anomaly triage is a natural home for a weak-signal architecture in which bounded findings raise attention first, candidate meanings remain provisional, and promotion to stronger operational significance occurs only through declared persistence, corroboration, and governance predicates. That is the narrow transfer claim of this bridge note.

The contribution is intentionally compact. It provides a minimal architecture contract for bounded findings, provisional candidate states, attentional escalation, and governed promotion, together with three short decision flows showing how the pattern applies to transformer stress precursors, demand-response ambiguity, and DER or microgrid instability precursors. The result is not a control framework or a utility operations doctrine. It is a disciplined architectural applicability note for handling uncertain grid-side signals without collapsing prematurely into either indifference or certainty.